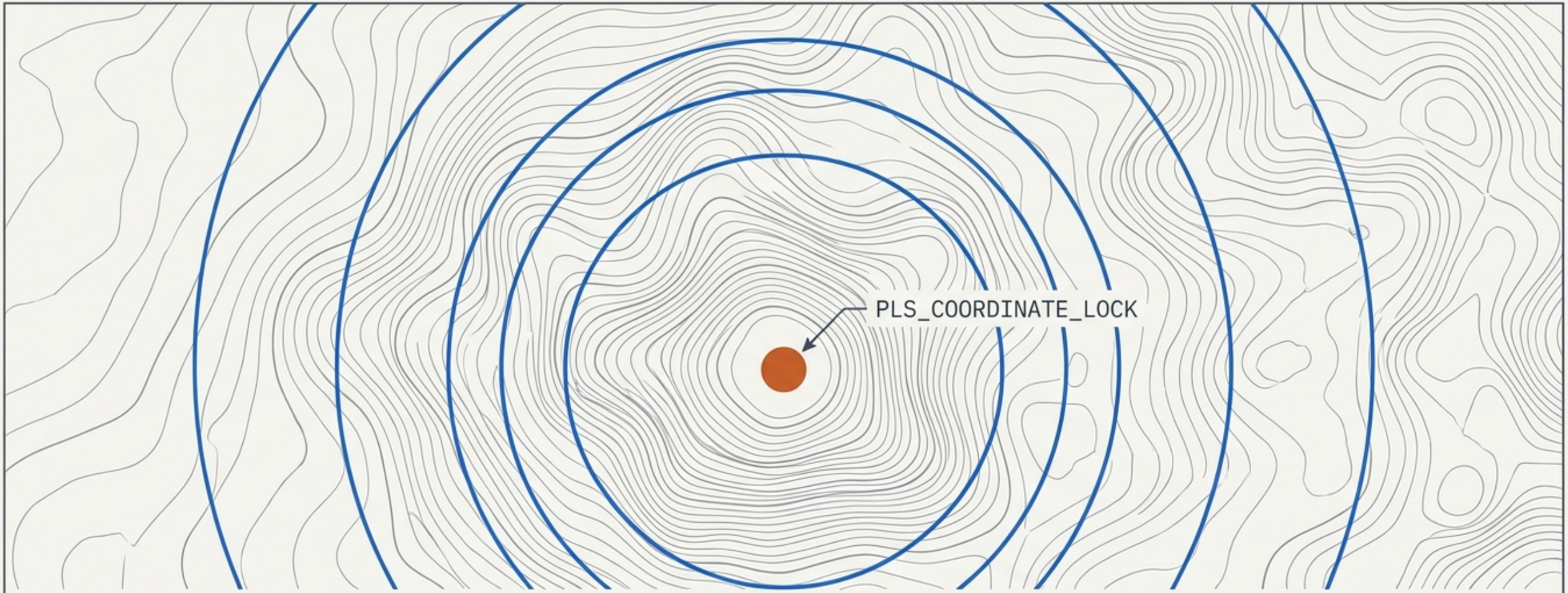

PROJECT SENTINEL: A RUST FRAMEWORK FOR MULTI-MODAL SEARCH & RESCUE ANALYSIS

Architecting an open-source, continuous-evaluation engine for missing persons in the US and Canada.

Synthesizing ISRID statistical models, eDNA forensics, and OSINT digital intelligence.

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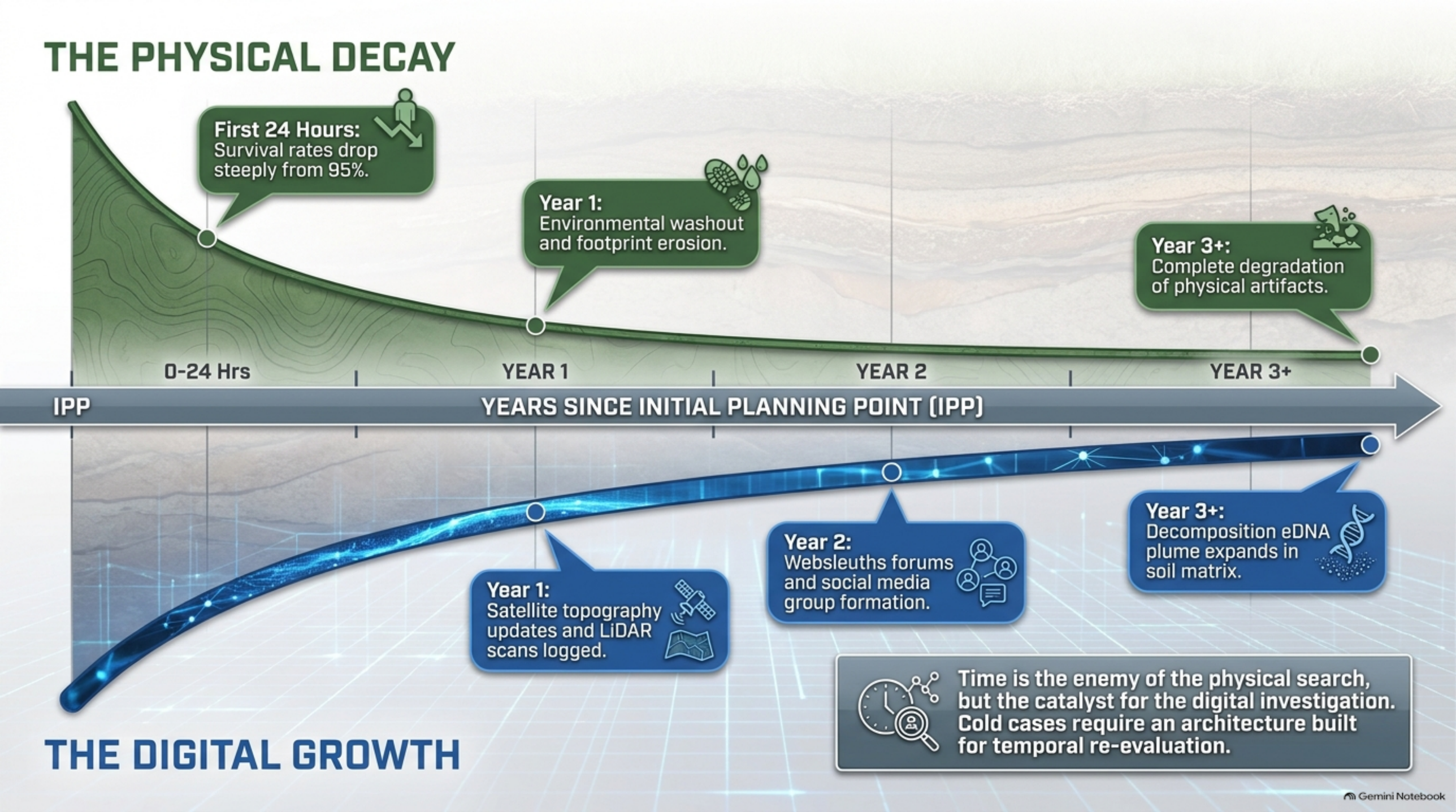

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SAR Operations

Relies on **physical grid sweeps, drone flight paths**, and **SARCAT records**. Evaluates **Cumulative Probability of Detection (POD)**.

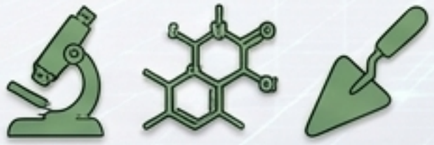
Statistical Models

Relies on the **International Search & Rescue Incident Database (ISRID)**, analyzing **143,951+ past incidents**.

UNSOLVED PLS

Forensic Science

Relies on **soil testing**, clandestine grave chemical plumes, and Locard's Exchange Principle.



OSINT / Digital

Relies on **cell tower forensics**, family trip plans, and Facebook/Reddit chatter.

DIAGNOSIS: Law enforcement, volunteer SAR, forensic labs, and OSINT investigators operate in disjointed ecosystems. Missing persons fall through the gaps between these incompatible systems.

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	Static SAR (Legacy)	Dynamic SAR (The Sentinel Framework)
Timeframe	One-time rapid deployment (focused heavily on the first 72 hours).	Continuous, remote year-over-year **algorithmic re-evaluation**.
Data Ingestion	Manual entry, paper-based Missing Person Questionnaires (MPQ), verbal reports.	**Automated API ingestion** (LiDAR point clouds, OpenDrift vectors, **continuous web scraping** of social chatter).
Analysis Engine	Subjective map drawing based on tabletop consensus and static radii.	**Deterministic Rust code** running thousands of **Monte Carlo simulations** and **MapScore** probability scoring.
Case Status	Cases freeze when the physical ground search is exhausted.	Cases **evolve automatically** as new **digital**, **social**, and **environmental data layers** are logged each year.

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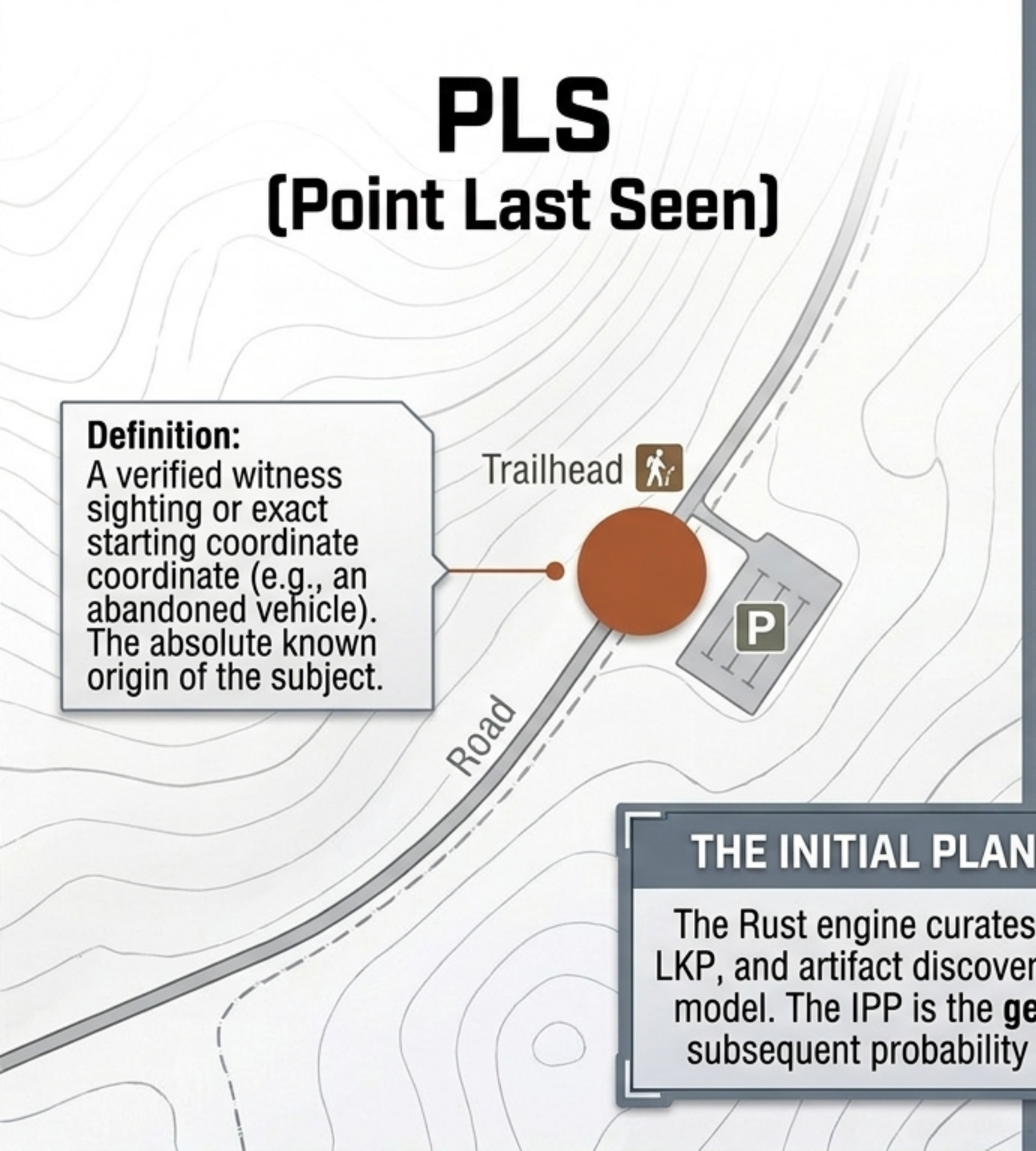
PLS
(Point Last Seen)

Definition:
A verified witness sighting or exact starting coordinate coordinate (e.g., an abandoned vehicle). The absolute known origin of the subject.



LKP
(Last Known Position)

Definition:
A clue-inspired location (e.g., a dropped artifact, a campsite, or a cell phone tower ping). An inferred location based on forensic or digital evidence.

THE INITIAL PLANNING POINT (IPP)

The Rust engine curates all GPS locations (PLS, LKP, and artifact discoveries) to anchor the spatial model. The IPP is the **genesis coordinate** for all subsequent probability rings and drift models.

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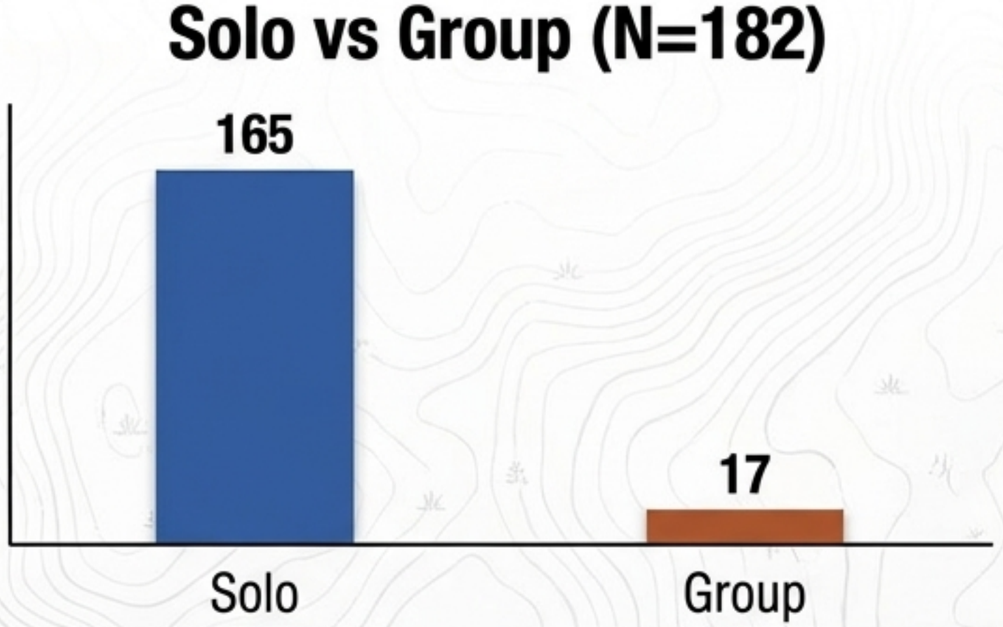
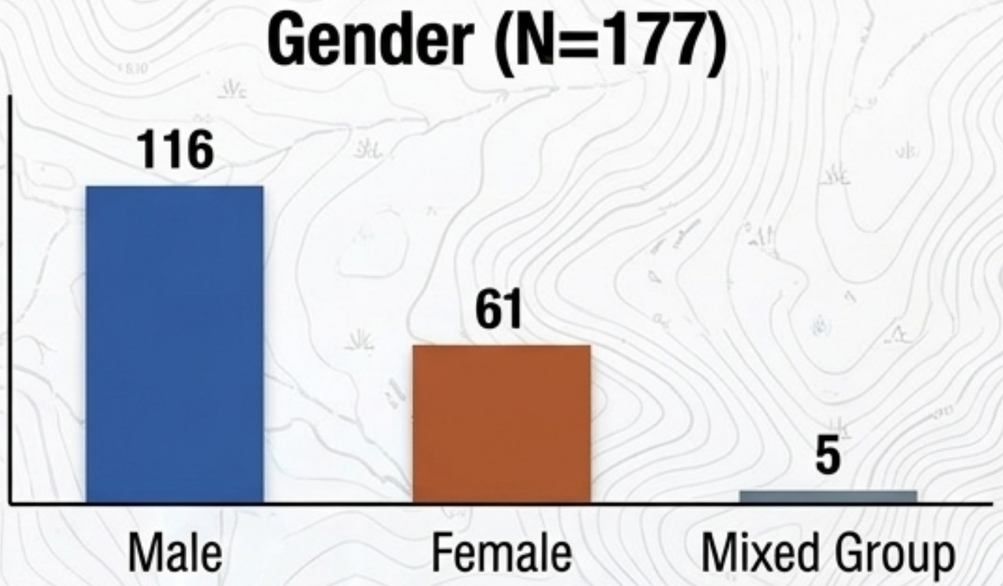
The ISRID Foundation: Lost Person Behavior

The Statistical Engine

Built on 143,951+ incidents, controlling for 41 subject categories (e.g., Dementia, Hiker, Child Wanderer), ecoregion, and terrain.

Calculates 25%, 50%, 75%, and 95% probability distance rings from the IPP.





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Anomalous Behavior: Breaking the Statistical Model

CASE 01: The Evasive / Fugitive

(e.g., Steven Stayner case)

Subjects actively avoiding detection break standard distance rings.
Requires automated cell-tower triangulation and social media monitoring to track evasive, non-Euclidean movement.

CASE 02: The Dissociative

(e.g., Agatha Christie, Brenda Heist)

Dissociative fugue states where subjects assume new identities or hitchhike long distances.

Statistical terrain models fail completely; long-term digital footprint tracking becomes mandatory.

CASE 03: The Abduction

(e.g., Jan Broberg, Colleen Stann)

Third-party actors move the subject. Distance traveled is dictated by the perpetrator's vehicle, not walking speed.

Relies heavily on vehicle telematics and OSINT anomalies.

SYNTHESIS:

When human behavior deviates from statistical probability, the Rust engine automatically **triggers OSINT web-scraping modules** to track digital echoes far beyond the physical search perimeter.

SYNTHESIS COMPLETE

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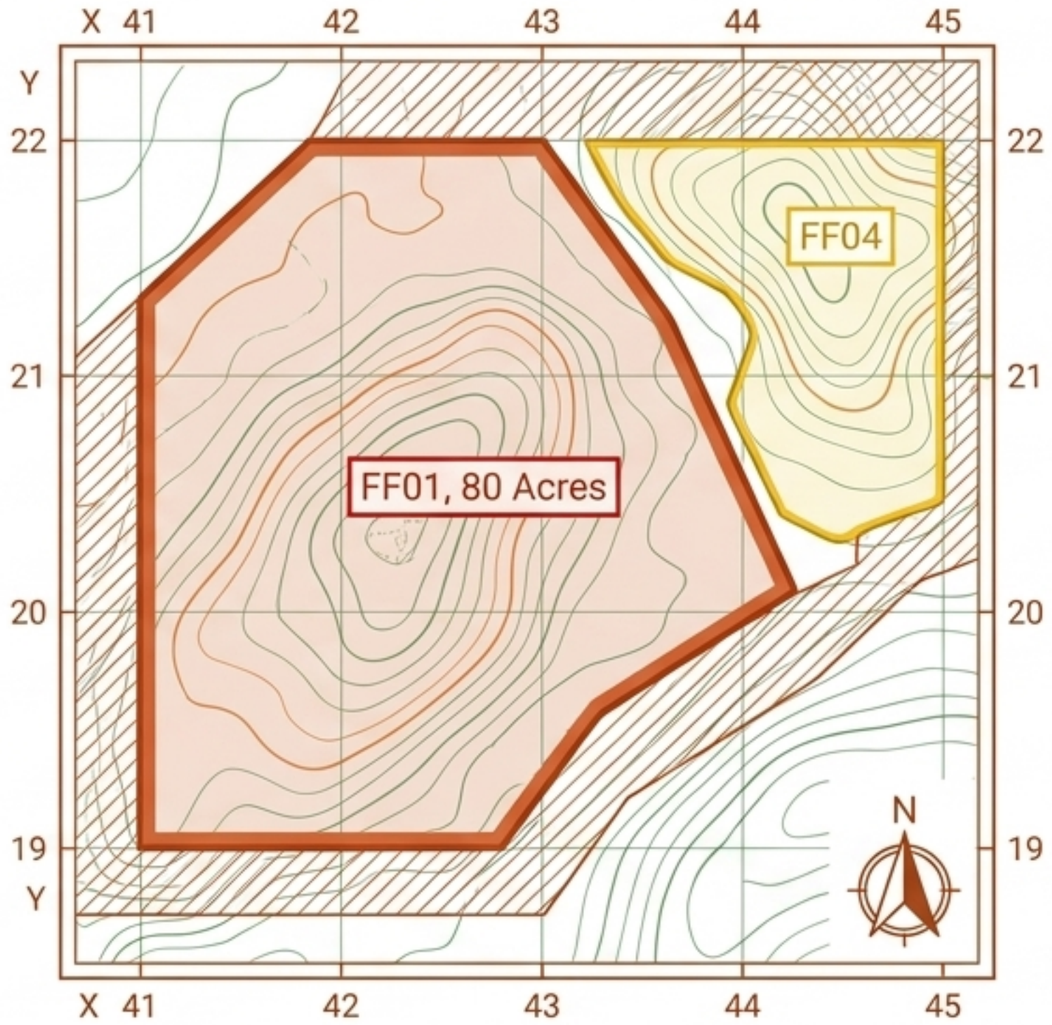
Coverage Path Planning (CPP) & MapScore

$$OPOS = \sum (POD \times POA)$$

Overall Probability of Success = Probability of Detection × Probability of of Area

The Sentinel framework integrates Charnes-Cooper and Washburn algorithms. It objectively evaluates Euclidean distance-rings against Watershed models, prioritizing sectors based on real-world terrain constraints.



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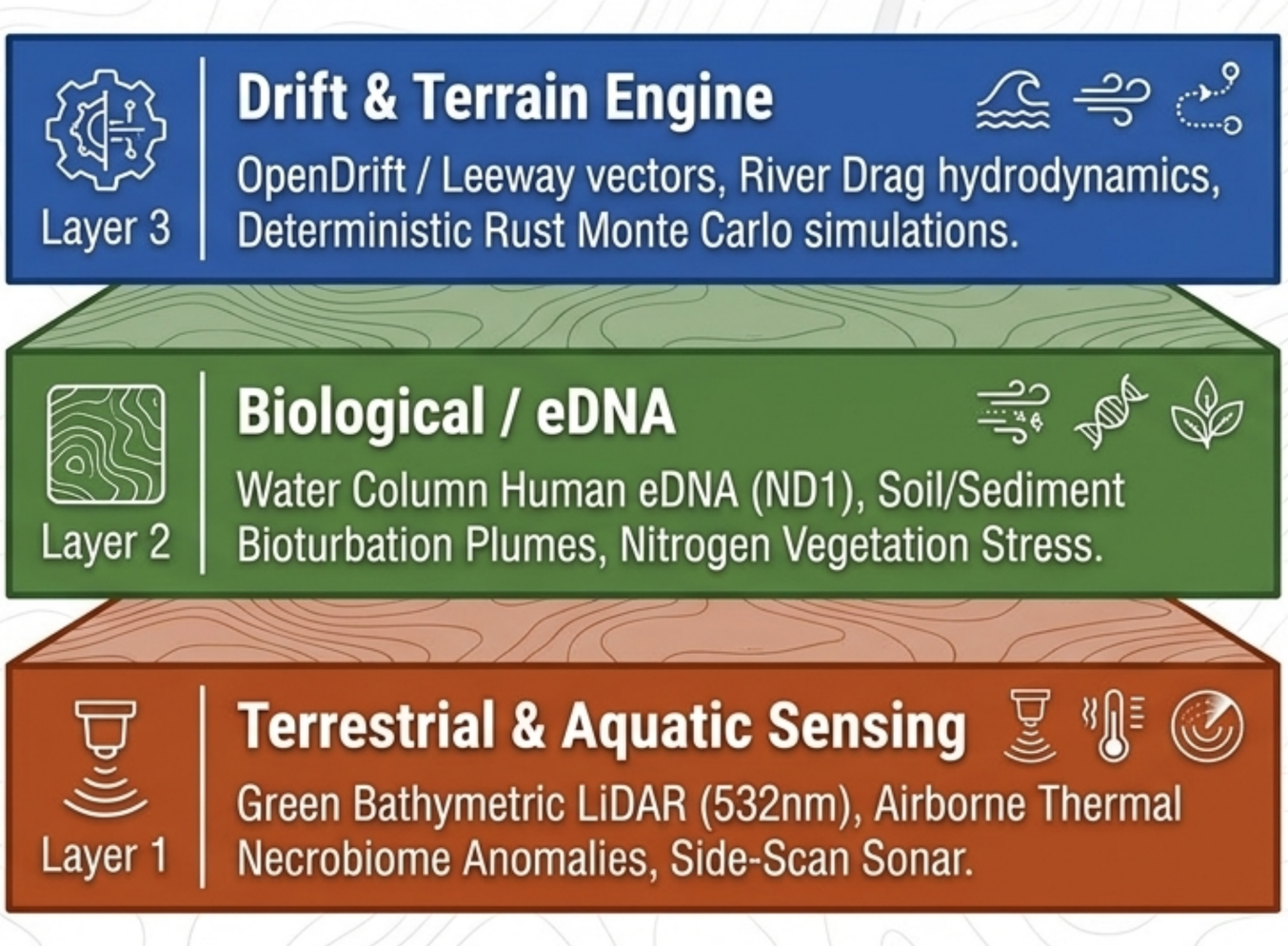
The Audrey Edna Framework (AEF) Adaptation

A 3-Tier Blueprint for Multi-Sensor Data Fusion



The 3-Tier Confirmation Scale

Level 1: Remote Sensor Anomaly

Level 2: Hydrodynamic/ eDNA Verification

Level 3: Physical Confirmation

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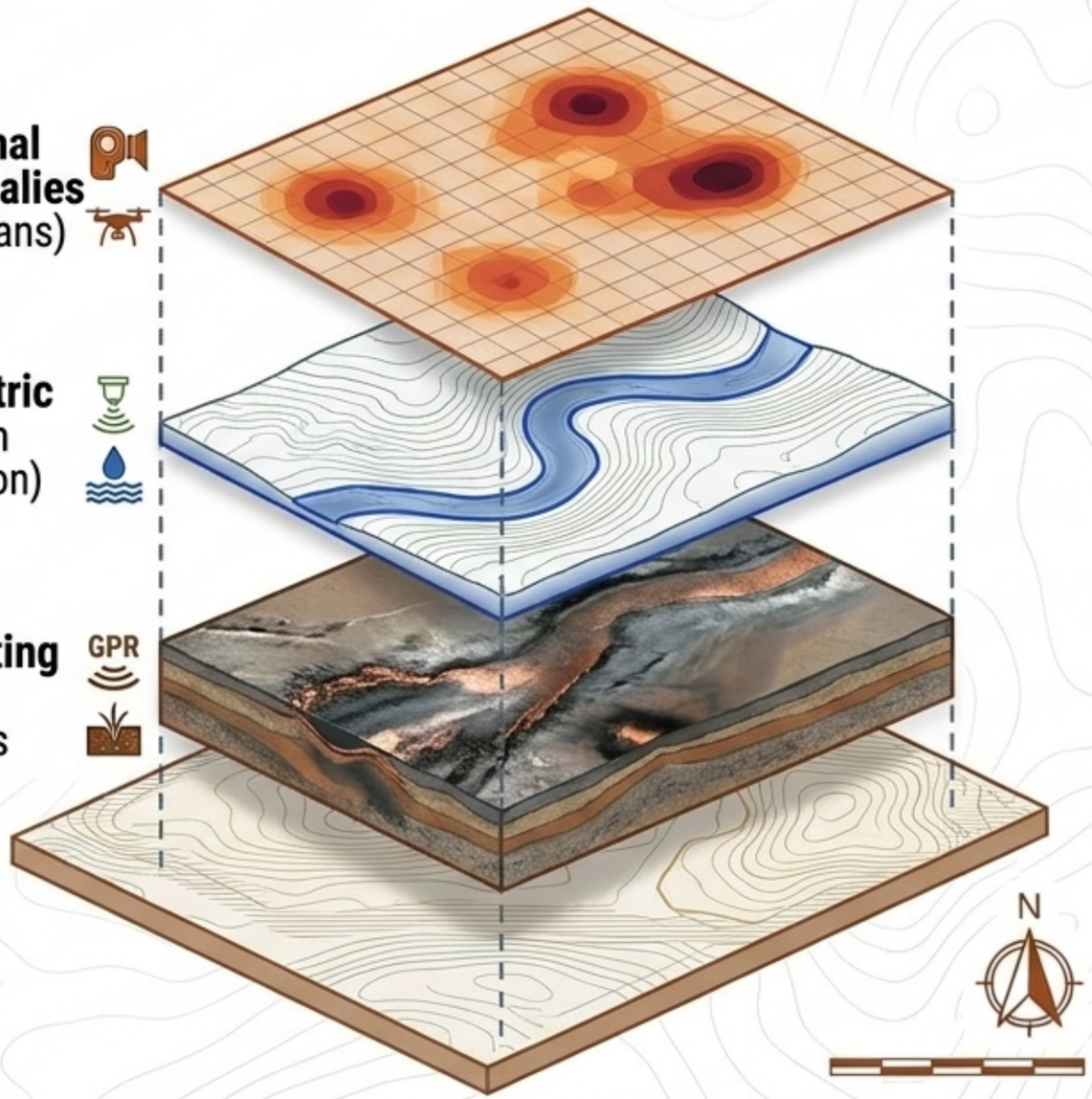
Terrestrial & Sub-Surface Sensors

Year-Over-Year Terrain Evaluation:

The Rust engine continually ingests new open-source satellite data. It applies auto-segmentation rules to update the Probability of Area (POA) as the landscape changes due to forest fires, erosion, or seasonal foliage shifts.



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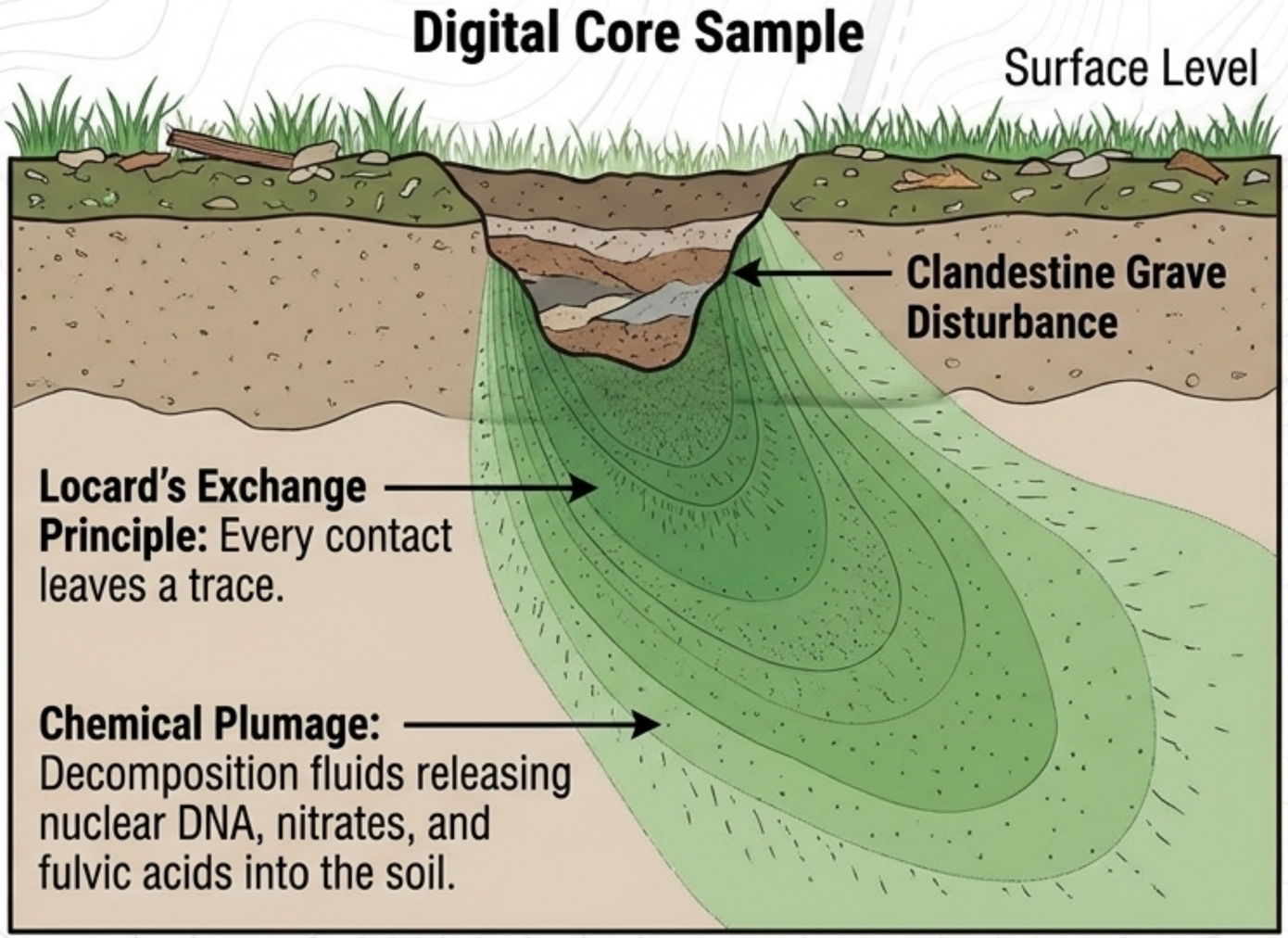
Forensic Soil Analysis: The Biological Signature



Lopez 2021 Extraction Analysis	
Extraction Method	**Human Nuclear DNA Yield**
Silica-based Extraction	0.0445 ng (High Inhibition)
Organic Extraction	1.6929 ng (Viable for STR Genotyping)

Even if a perpetrator relocates a victim, the repository's eDNA modules map the residual chemical plume at the initial site, tracking degradation over years.

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Aquatic Hydrodynamic Drift Models

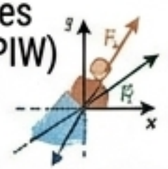
Shifting from Terrestrial Probability to Fluid Dynamics

OpenDrift & USCG Leeway

Incorporating MET Norway's open-source Python framework into the backend for coastal SAR.

The Physics of Leeway

Algorithms calculating Lagrangian particle trajectories based on Persons in Water (PIW) immersion levels, windage, and wave radiation forces.



PMSI & Backtracking

Utilizing Postmortem Submersion Interval gas-refloat equations and reverse time-stepping to backtrack floating artifacts to their origin point (IPP).

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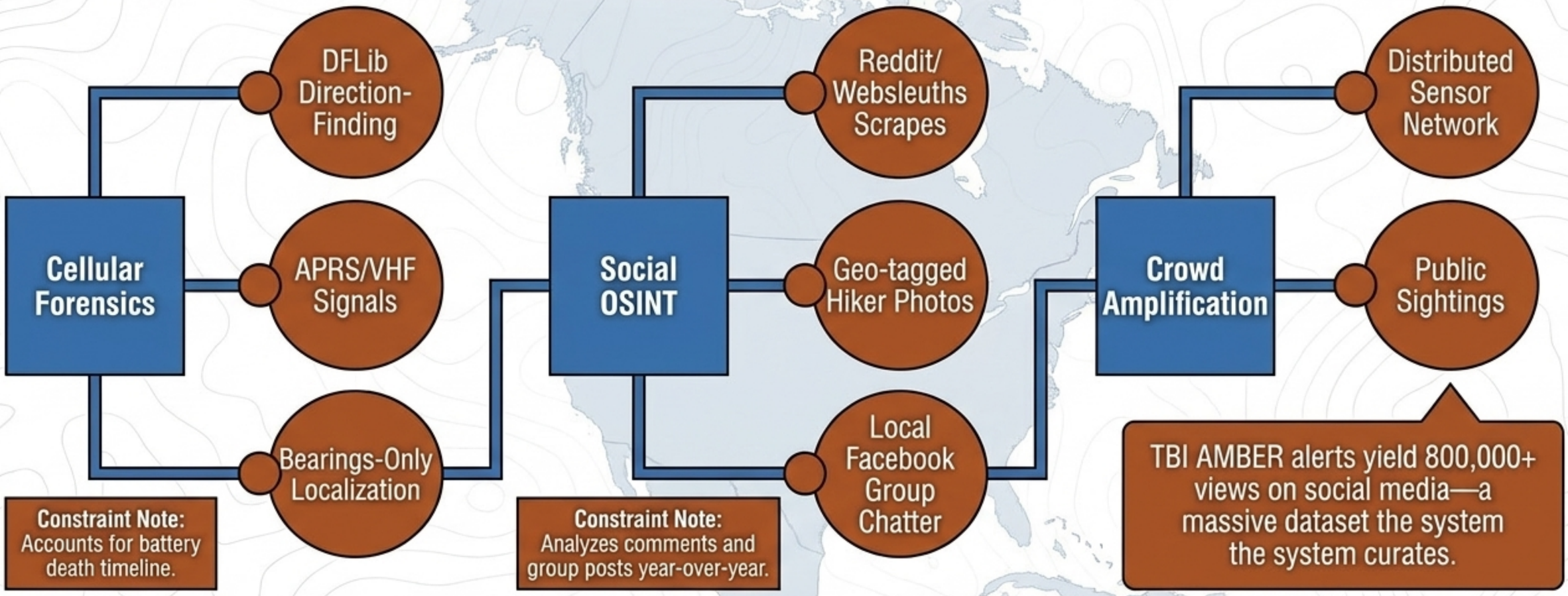
The Digital Echo: OSINT & Social Telemetry



The Rust engine scrapes, curates, and logs digital echoes over each year a person is missing, finding the last point of sight or artifact left behind.

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Synthesis: The Digital Core Sample

Looking through all data layers simultaneously at a single GPS coordinate

Solving cold cases requires vertical synthesis.

The Rust engine correlates an OSINT photo with a **LiDAR anomaly, ground detection radar**, and an **eDNA soil test** to generate a **unifate a unified, high-probability search sector** for multi-year cases.

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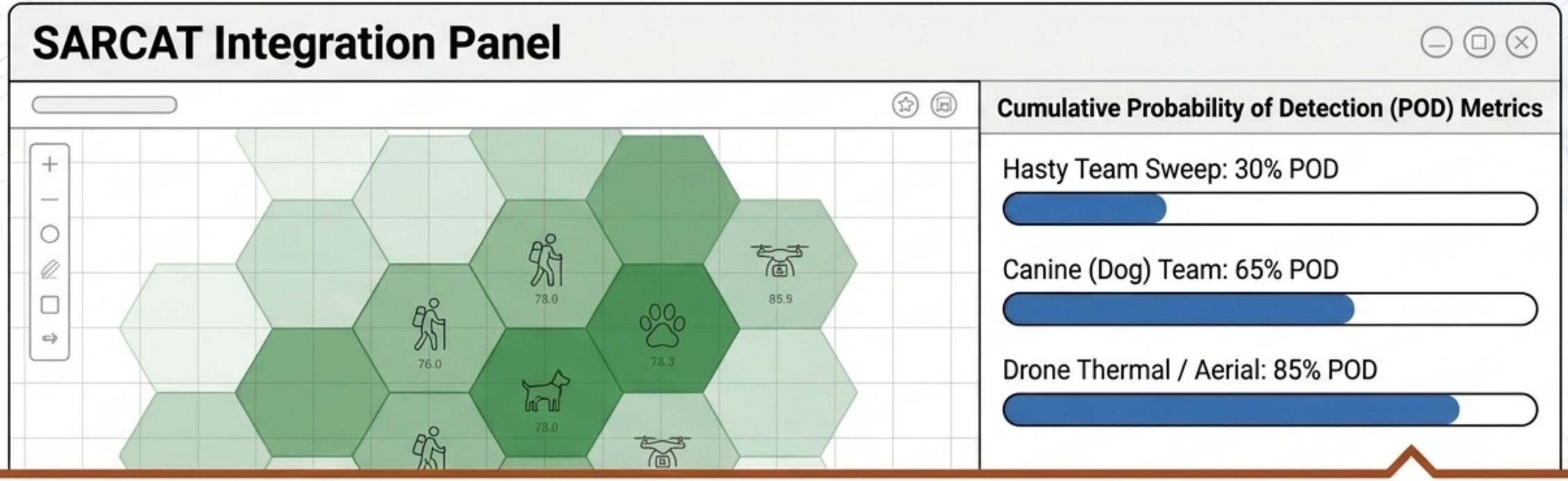
SARCAT & Cumulative POD

Tracking exactly what tests were done and what areas were cleared.



THE COLD CASE RULE: Find out where the person is NOT.

To remotely evaluate a case years later, the system must log exactly what tests were done, what algorithms were used, and how thoroughly the sector was cleared initially. **We must avoid redundant efforts.**

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System Architecture: Why Rust?

```
fn concurrency()
```

Concurrency & Performance

Capable of running thousands of deterministic Monte Carlo drift simulations and Charnes-Cooper algorithms simultaneously without bottlenecking.

```
fn memory_safety()
```

Memory Safety

Handles massive, volatile geospatial datasets (unstructured PDFs, point clouds, raster maps) without memory leaks or crashes during critical processing loops.

```
fn process_lidar()
```

Unmatched Speed

The optimal language for parsing millions of data points in Bathymetric LiDAR, ground detection radar, and thermal drone imagery in near real-time.

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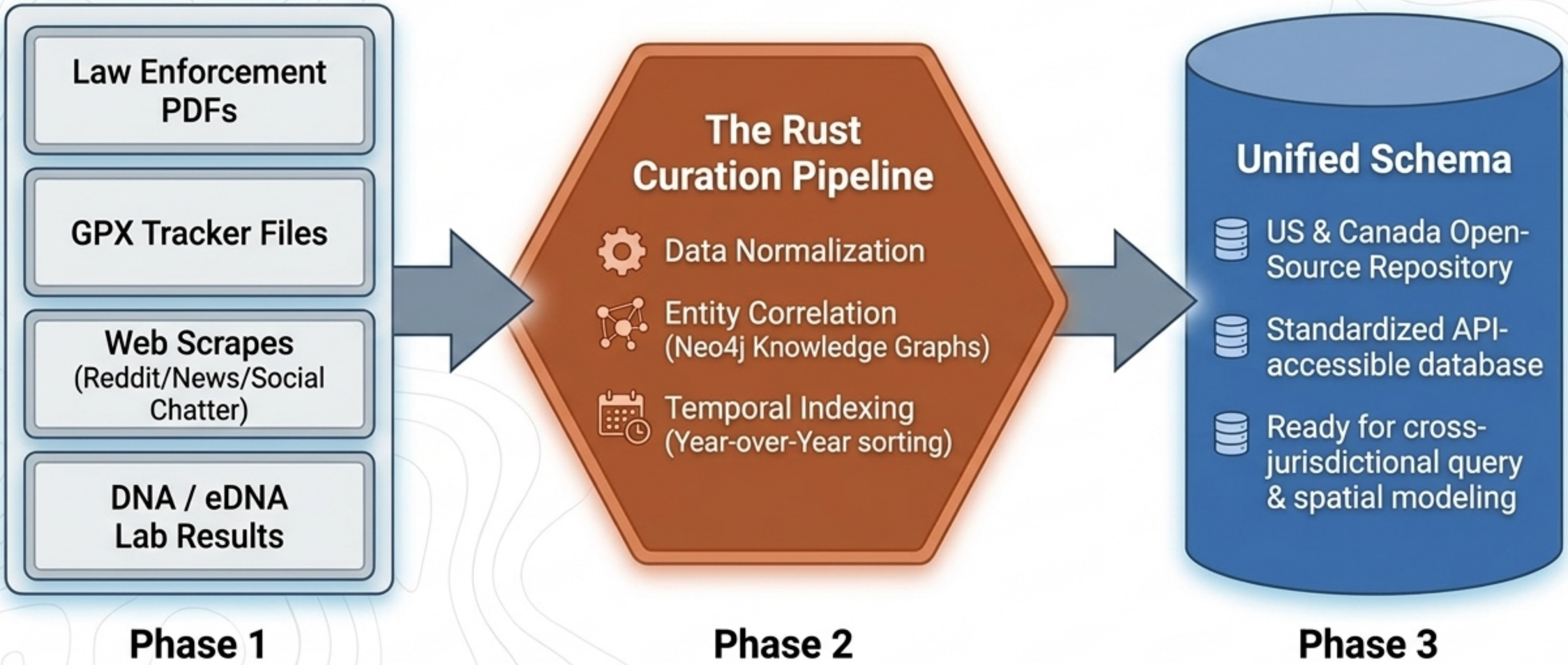
Ingestion & Curation Pipeline



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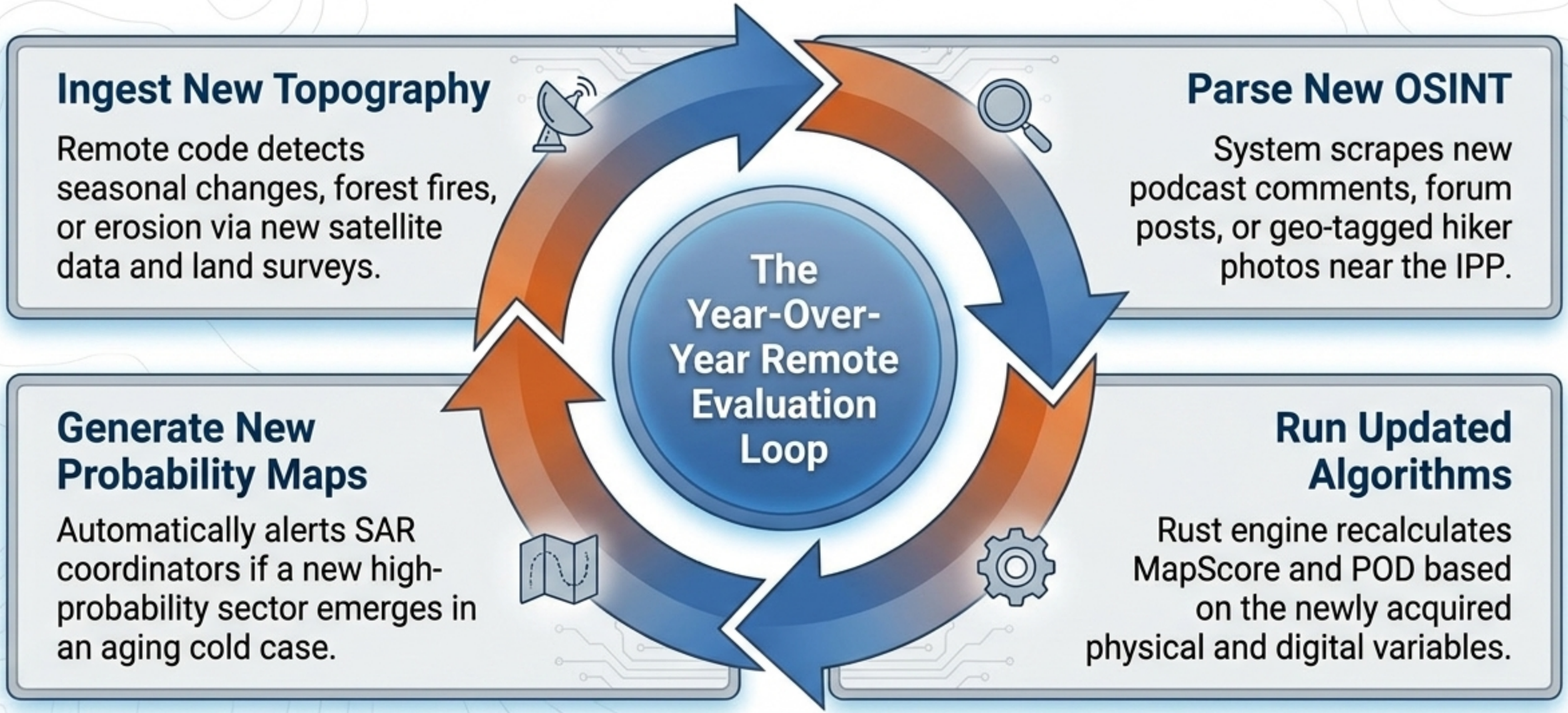
The Temporal Investigation Architecture



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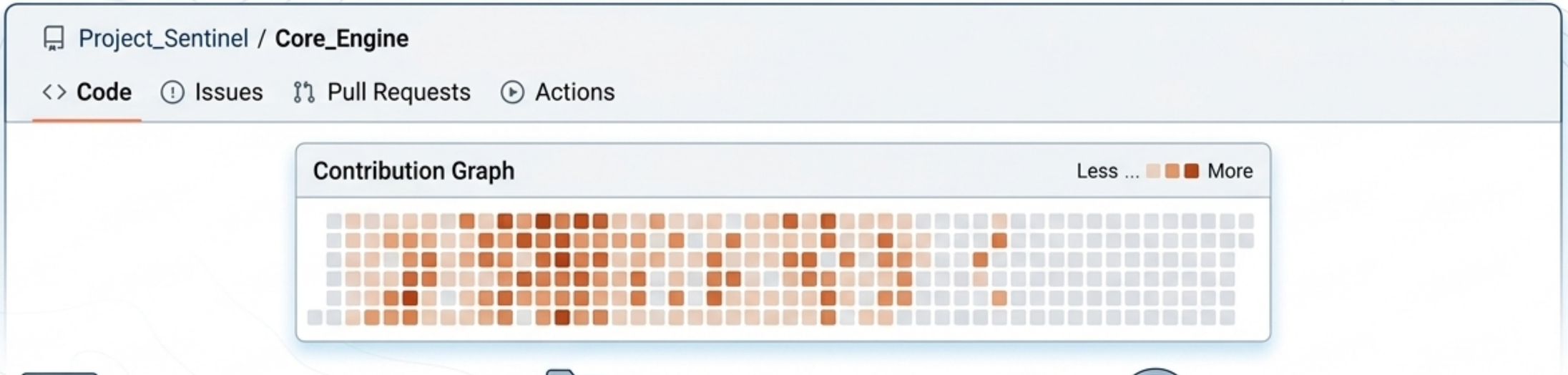
Building the US & Canada Repository

A Call to Action for the Open-Source Community



For Developers

Contribute Rust modules for OSINT scraping, eDNA API integration, and LiDAR parsing.

For SAR Coordinators

Begin digitizing historical case files via SARCAT for schema alignment and lost persons databases.

For Investigators

Utilize the open-source repo to run legacy cold cases through the new deterministic models.

A search is an unknown problem at an unknown location. We build the architecture to make the unknown visible—one layer, one year at a time.

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